

Programming Fundamentals

Deadline: October 5, 2025

Course Instructor

Ms. Anosha Khan

Course Teacher Assistant

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Assignment 2

Total Time (Days): 7

Total Marks: 100

Total Questions: 8

Instructions to be followed

Include comments in your code wherever necessary to improve clarity. The use of Artificial Intelligence tools or Large Language Models (LLMs), such as ChatGPT or DeepSeek, is strictly forbidden. Any student found using AI or LLMs to complete this assignment will receive **zero marks** and may face **serious disciplinary action (DC)** in the future.

If you are unsure about any part of the assignment, consult your **TA for clarification** before making assumptions.

Submission Instructions

Solve each question in a **separate C++ file**. Name each file using this format (RollNumber_Q1.cpp, RollNumber_Q2.cpp) Example: 25L_XXXX_Q1.cpp, 25_XXXX_Q2.cpp and so on. Put all your C++ files into **one ZIP file** with the following format (**25L_XXXX.zip**) and submit that **ZIP File** on Google Classroom.

Important: Submissions that do not follow these instructions will result in **deduction of marks**. It is your duty to submit working code on GCR following the format given above. Students are expected to ensure that their programs **handle edge cases** with a user-friendly output if needed. Also handle **unwanted values & errors**.

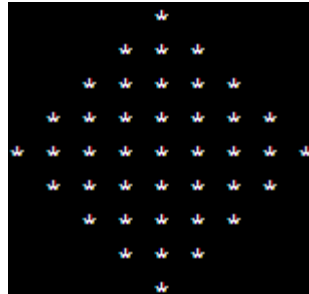
The syllabus includes Loops (for, while, do...while) and topics covered before this. The use of advanced topics will lead to deduction of marks.

Flip the Page for Questions

Question 01

Write a Program that Takes an Input **N** from User. Here, **N** is the Number of rows to print

Example: For N = 5, Following Pattern should be printed.



Question 02

Write a Program that Takes Input N which is the Number of rows to print the following pattern e.g. For N = 5, Following Pattern should be printed.



Question 03

Write a Program that takes Input N. Here, N must be a Perfect Square e.g. 1,4,9,16,25,36 etc. Following Pattern should be printed e.g. For N=25

1	2	3	4	5
16	17	18	19	6
15	24	25	20	7
14	23	22	21	8
13	12	11	10	9

Question 04

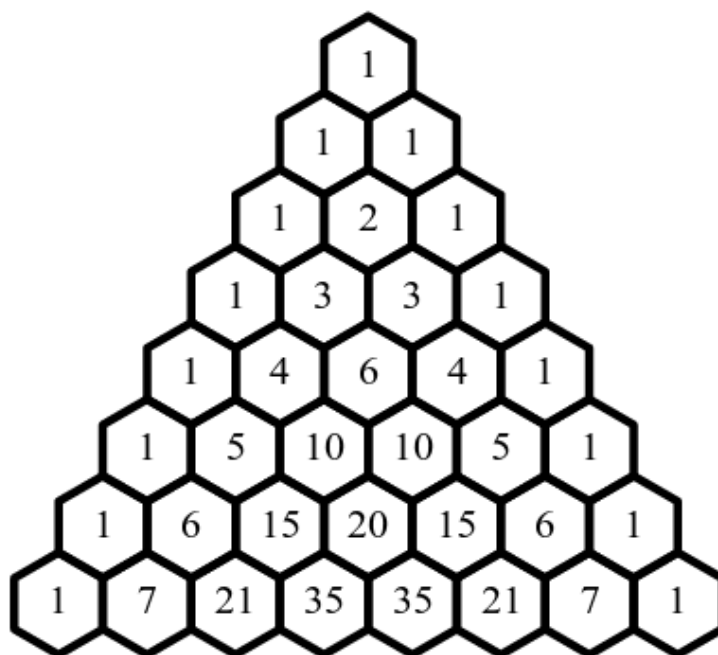
Write a program that asks the user for Number of Courses (N) he/she is taking in a semester. Then take the grade points and credit hours for each course as input, then calculates and displays the SGPA (Semester Grade Point Average).

$$\frac{\sum_{i=0}^N (Grade\ Point)^i \times (Credit\ Hours)^i}{\sum_{i=0}^N (Credit\ Hours)^i}$$

- $(Grade\ Point)^i$ represent the expected Grade points in ith Course (A+ 4.00, A 4.00, A- 3.67 ...)
- $(Credit\ Hours)^i$ represent the Credit Hours of ith Course (3,2,1)

Question 05

Write a Program that Prints a Pascal Triangle. The Number of Rows N should be taken from user e.g. For N = 8, The Following Pattern should be printed



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Question 06

A coffee shop named “**FAST Hub**” sells three types of coffee: **Espresso**, **Cappuccino**, and **Latte**. The prices are as follows:

- Espresso = Rs. 230
- Cappuccino = Rs. 250
- Latte = Rs. 150

The user will enter the type of coffee they want and the number of cups. Also, ask if the customer wants **extra sugar (Rs. 30)** or **whipped cream (Rs. 50)**. Finally, calculate the total bill with **20% service tax**, and display the complete order details. In this Program, you will be creating a Menu that Displays the UserID, The Total Price and GST at the end. Your Program should ask the user as many times he want to order, and as many cups as Want. After each Order, you should be Displaying the Subtotal Bill with GST as well along with Order Details. UserID should increment after each order.

Question 07

Write a program that generates the multiplication table of a given number. The program should first ask the user to enter the number for which they want the table. It should then ask how far the table should go. Finally, the program should display the multiplication table of the chosen number, starting from one and continuing up to the given limit. For example, if the user enters 10 as the number and 20 as the limit, the output should display the table of 10 from $10 \times 1 = 10$ up to $10 \times 20 = 200$.

Question 08

Write a program that takes N Number and print N prime numbers e.g. N=5 The Following Numbers should be printed 2,3,5,7,11

I hope you all will learn something new
while doing this Assignment
Best of Luck Guys
See you all energetic than before in Evaluation
Best Wishes